AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M218
Relational databases

Final Examination
Spring Semester 2024/2025

Date: 28 /May/2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks:50
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I [10 Marks]

This part includes 5 MCQ; each weights 2 marks. Choose the most suitable answer for the following questions:

1. What does an Entity-Relationship (ER) diagram represent?
   a) The hardware components of a database system
   b) The relationships between tables in a database
   c) The interaction between a database and its users
   d) The indexing strategy for optimizing queries

2. Which of the following best describes a weak entity?
   a) An entity that does not have a primary key on its own
   b) An entity that has no attributes
   c) An entity that cannot participate in relationships
   d) An entity that does not store any data

3. Which of the following is NOT a valid type of relationship in an ER model?
   a) One-to-One (1:1)
   b) One-to-Many (1:N)
   c) Many-to-Many (M:N)
   d) None-to-One (0:1)

4. Which relational algebra operation removes duplicate records from a result set?
   a) SELECTION (σ)
   b) PROJECTION (π)
   c) UNION ($\cup$)
   d) INTERSECTION ($\cap$)

5. How does an M:N (Many-to-Many) relationship get represented in a relational database?
   a) By creating an additional table that contains foreign keys from both participating tables
   b) By directly linking two tables without an intermediary table
   c) By duplicating rows in each table to store the relationship
   d) By using a self-referential foreign key

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Part II [40 Marks]

Question 1 [15 Marks]

You have been hired to design a database for a boat training school that offers boating courses. The school keeps track of: Students who enroll in courses, Boat Courses that students can take, Enrollments that link students to the courses they registered for.

Your task is to design a database for a boat training school. The database will store details about students, the boat courses they enroll in, and their enrollment records.

Using the given details, you need to write the appropriate SQL commands to create these tables while ensuring that all constraints are correctly implemented.

Students Table

The Students table stores information about individuals who enroll in boat training courses.

- Each student should have a unique identifier to distinguish them from others.
- The student's name must be stored and cannot be left empty.
- The student's age must be at least 18 years old.
- Each student has an experience level which can be either Beginner, Intermediate, or Advanced. If not specified, it should default to "Beginner".

BoatCourses Table

The BoatCourses table contains information about the different courses offered at the boat training school.

- Each course should have a unique identifier.
- The course name must be provided and cannot be left empty.
- The duration of the course should be recorded in weeks and should be between 1 and 10 weeks.
- Each course is taught by a specific instructor, whose name should be stored.

Enrollments Table

The Enrollments table keeps a record of which students have registered for which courses.

- Each enrollment should have a unique identifier.
- Each enrollment must be linked to a specific student, meaning it should reference the Students table.
- Each enrollment must also be linked to a specific course, meaning it should reference the BoatCourses table.

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- The date of enrollment should be stored. If no date is provided, it should default to the current date.

1. Create the database and tables as described above. Write the SQL DDL statements to create the Students, BoatCourses, and Enrollments tables. **[6 marks]**

2. Insert sample data into the tables **[3 marks]** .

- Insert the following records:
  - *Student Table:*
    - *(1, 'Omar Khalid', 25, 'Intermediate')*
    - *(2, 'Layla Ahmed', 30, 'Beginner')*
  - *BoatCourses Table:*
    - *(101, 'Basic Sailing', 4, 'Captain Ali')*
    - *(102, 'Advanced Navigation', 6, 'Captain Zaid')*
  - *Enrollment Table:*
    - *(1, 1, 101, '2024-07-01')*
    - *(2, 2, 102, '2024-07-02')*

3. Modify the Students table to add a new column for Email **[2 marks]**.

- The email should be unique and cannot be NULL.

4. Update the experience level of student Layla Ahmed to Advanced **[2 marks]**.

5. Delete the boat course with CourseID = 102 from the BoatCourses table **[2 marks]**.

Question 2 [25 Marks]

You are managing a library book rental system where members can borrow books. You need to write SQL queries to retrieve and manipulate data from the system.

Assume the following tables exist:

```
CREATE TABLE Members (
    MemberID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Age INT CHECK (Age >= 12),
    MembershipType VARCHAR(50) DEFAULT 'Standard'
);

CREATE TABLE Books (
    BookID INT PRIMARY KEY,
    Title VARCHAR(200) NOT NULL,
    Author VARCHAR(100),
    Genre VARCHAR(50),
    AvailableCopies INT CHECK (AvailableCopies >= 0)
);
```


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```
CREATE TABLE Rentals (
    RentalID INT PRIMARY KEY,
    MemberID INT REFERENCES Members(MemberID),
    BookID INT REFERENCES Books(BookID),
    RentalDate DATE DEFAULT CURRENT_DATE,
    ReturnDate DATE NULL
);
```

1. Retrieve all books in the library, displaying their title, author, and genre **[2 marks]**.
2. Find the names and membership types of all members who are older than 18 **[2 marks]**.
3. List all rentals made on or after January 1, 2024 **[2 marks]**.
4. Retrieve the titles of books that have more than 2 available copies **[2 marks]**.
5. Show all rentals that have not been returned yet (ReturnDate is NULL) **[2 marks]**.

6. Find the number of books rented by each member **[3 marks]**.
7. List the members who have rented the book titled "To Kill a Mockingbird." **[3 marks]**.
8. Retrieve the most popular book (the book that has been rented the most times). **[3 marks]**.
9. List all books that have never been rented. **[3 marks]**.
10. Find the members who have rented books from both the 'Fiction' and 'Science' genres. **[3 marks]**.

End of questions

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